

Review of linear, homogeneous ODEs (Chapter 2)

$$(*) \quad \dot{X} = A(t)X$$

$$X \in \mathbb{R}^n$$

$A(t) = n \times n$ matrix
(continuous)

$X=0$ is a solution (homogeneous)

if $X_1(t)$ & $X_2(t)$ satisfy $(*)$, then so does $c_1 X_1(t) + c_2 X_2(t)$ for any c_1, c_2
(linear superposition principle)

If we can find n linearly independent solns. $X_1(t), X_2(t), \dots, X_n(t)$ then we have found the general soln to $(*)$, which can be written as

$$X(t) = \sum_{k=1}^n c_k X_k(t)$$

Special case $A = \text{constant matrix}$ (autonomous)

- Approach 1

Compute eigenvalues λ of A as roots of the characteristic eqn. $\text{Det}(A - \lambda I) = 0$.

If none are repeated then we know we can write the general soln. as

$$X(t) = c_1 e^{\lambda_1 t} V_1 + c_2 e^{\lambda_2 t} V_2 + \dots + c_n e^{\lambda_n t} V_n$$

where $v_1, \dots, v_n$ are linearly independent eigenvectors $Av_j = \lambda_j v_j \quad j=1, \dots, n$

ie.
$$X(t) = \begin{bmatrix} | & | & | \\ e^{\lambda_1 t} v_1 & e^{\lambda_2 t} v_2 & \dots & e^{\lambda_n t} v_n \\ | & | & | \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

initial condition $X(0) = X_0$ determines $\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$

ie.
$$X_0 = \begin{bmatrix} | & | & | \\ v_1 & v_2 & \dots & v_n \\ | & | & | \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} | & | & | \\ v_1 & \dots & v_n \\ | & | & | \end{bmatrix}^{-1} X_0$$

← guaranteed to be invertible since the v 's are linearly independent

$$\Phi(t) = \begin{bmatrix} | & | & | \\ e^{\lambda_1 t} v_1 & e^{\lambda_2 t} v_2 & \dots & e^{\lambda_n t} v_n \\ | & | & | \end{bmatrix} = \text{"Fundamental matrix soln"}$$

$\Phi(t)$ satisfies

$$\dot{\Phi} = A \Phi$$

[nice convention to add to this $\Phi(0) = \text{Id}$
 then soln. to $\dot{X} = A X$, $X(0) = X_0$ is
 $X = \Phi(t) X_0$]

aside:

If any eigenvalue λ is complex, i.e. $\lambda = \alpha + i\beta$,
 then $\lambda^* = \alpha - i\beta$ is also an eigenvalue

$$A v = \lambda v, \quad A v^* = \lambda^* v^*$$

and we can construct a real soln. from
 these as

$$e^{\lambda t} v + c^* e^{\lambda^* t} v^*, \quad \text{where } c \in \mathbb{C}$$

and this can be written as

$$e^{\alpha t} (a \cos(\beta t) + b \sin(\beta t))$$

or

$$e^{\alpha t} (A \cos(\beta t - \phi))$$

Note if $\alpha = \text{Re}(\lambda) > 0 \Rightarrow$ exponentially growing
 $\alpha < 0 \Rightarrow$ exponential decay

What about repeated eigenvalues?

Then it could be that it's a defective
(matrix ~~can't~~ be diagonalized)

$$\underbrace{\text{geometric multiplicity}}_{\text{dim. of eigenspace}} < \text{algebraic multiplicity}$$

$\Rightarrow$ not enough eigenvectors to construct n
linearly independent solns as $e^{\lambda t} v$

Additional solns. obtained in terms of
"generalized eigenvectors" & have t -dependence of
form $t e^{\lambda t}$, $t^2 e^{\lambda t}$, ...

example: v is an eigenvector of A
with eigenvalue λ of multiplicity 2.

$$(A - \lambda I)v = 0$$

w is a generalized eigenvector

$$\left. \begin{array}{l} (A - \lambda I)^2 w = 0 \\ (A - \lambda I)w \neq 0 \end{array} \right\} \Rightarrow (A - \lambda I)w = v$$

ie $Aw = v + \lambda w$

2 linearly independent solns are

$$x_1 = e^{\lambda t} v, \quad x_2 = e^{\lambda t} (tv + w) \quad \leftarrow \text{check this}$$

if multiplicity 3 $\Rightarrow x_3 = e^{\lambda t} \left(\frac{t^2}{2} v + tw + u \right)$

$\swarrow$ $(A - \lambda I)v = 0$ $\uparrow$ $(A - \lambda I)^2 w = 0$ $\nwarrow$ $(A - \lambda I)^3 u = 0$

Approach 2

$$\dot{X} = AX, \quad X(0) = X_0$$

has soln $X(t) = e^{At} X_0$

matrix

$$e^{At} = I + At + \frac{1}{2} A^2 t^2 + \frac{1}{3!} A^3 t^3 + \dots$$

$$= \sum_{k=0}^{\infty} \frac{A^k t^k}{k!} \quad (0! = 1)$$

converges for all t

Check claims:

$$X(0) = e^{A \cdot 0} X_0 = I X_0 = X_0 \quad \checkmark$$

$$\frac{dX}{dt} = \lim_{h \rightarrow 0} \frac{e^{A(t+h)} X_0 - e^{At} X_0}{h} = \lim_{h \rightarrow 0} \frac{X(t+h) - X(t)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^{At} e^{Ah} X_0 - e^{At} X_0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(e^{Ah} - I) e^{At} X_0}{h} = \lim_{h \rightarrow 0} \underbrace{\frac{(Ah + \frac{1}{2} A^2 h^2 + \dots)}{h}}_A X(t)$$

✓

Important : I needed $e^{A(t+h)} = e^{At} e^{Ah}$
 $= e^{Ah} e^{At}$

Is it true that $e^A e^B \stackrel{?}{=} e^{(A+B)}$ for matrices?

not necessarily — Baker-Campbell-Hausdorff Formula

$$e^A e^B = e^C, \text{ where}$$

$$C = A+B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \dots$$

$$\text{where } [A, B] = AB - BA$$

$$e^A e^B = e^{A+B} \text{ if } [A, B] = 0, \text{ i.e. } A \& B \text{ commute.}$$

in above I had $e^{A(t+h)} = e^{At} e^{Ah}$, which is justified since $[At, Ah] = th[A, A] = 0$

How do we evaluate e^{At} without having to sum all the matrices ---

Simple case: complete set of eigenvectors $v_1, \dots, v_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$
 (A is diagonalizable)

$$A = P \Lambda P^{-1} \quad (P^{-1} A P = \Lambda)$$

$$\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}, \quad P = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix}$$

P is invertible since v_j are linearly independent for complete set.

$$e^{At} = e^{P \Lambda P^{-1} t}$$

$$P \Lambda P^{-1} P \Lambda P^{-1} = P \Lambda^2 P^{-1}$$

$$= I + P \Lambda P^{-1} t + \frac{1}{2} (P \Lambda P^{-1})^2 t^2 + \dots$$

$$= P \left(I + \Lambda t + \frac{1}{2} \Lambda^2 t^2 + \dots \right) P^{-1}$$

$$\begin{pmatrix} 1 + \lambda_1 t + \dots & & \\ & 1 + \lambda_2 t^2 + \dots & \\ & & \ddots \end{pmatrix} = \begin{pmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & \ddots e^{\lambda_n t} \end{pmatrix}$$

$$= P \begin{pmatrix} \swarrow \\ \end{pmatrix} P^{-1}$$

But what if A is defective so we can't diagonalize it?

e.g. $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one eigenvector for eigenvalue $\lambda=1$, which is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Thm. (Proved in Chapter 2)

The matrix A on a complex space E has a unique decomposition $A = S + N$, where S is semi-simpl, N is nilpotent, and $[S, N] = 0$.

semi-simpl : $P^{-1}SP = \Lambda = \text{diagonal}$

nilpotent : $N^k = 0, N^{k-1} \neq 0$ nilpotency k
($k=1$ is the diagonalizable case)

ex. $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \Rightarrow S = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$$N^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

nilpotency 2

$$e^{tA} = e^{t(S+N)} = e^{tS} e^{tN} \quad \text{since } [S, N] = 0$$

$$e^{tN} = I + tN + \frac{1}{2}t^2N^2 + \dots$$

$$+ \frac{1}{(k-1)!} t^{k-1} N^{k-1}$$

Finite sum since

$$N^k = 0$$